

Adapttime: Method Overview

Thesis project

Current protocol

Abstract

Adapttime studies whether small fitted adaptors can combine a time-series foundation model with calendar-aligned retrieval while preserving the official TIME test protocol. This note defines the fixed validation-selected family and the causal rolling per-variate horizon Ridge; it reports no empirical result.

1 Chronological protocol

Each dataset and forecast term is divided into a fixed datastore, adaptation training, adaptation validation, and the untouched official TIME test interval. The first three regions end strictly before the test. Ridge candidates are fit once on adaptation training, K and α are selected on adaptation validation, and the selected coefficients are frozen for test. There is no train-plus-validation refit.

Let H be the forecast horizon, T_{test} TIME’s configured test length, and $n = \lfloor T_{\text{test}}/H \rfloor$ its official complete test-window count per variate. Adaptation training contains $2n$ window origins and validation contains n , giving three times the test-window count before test. Their interval lengths are $H + (2n - 1)s_f$ and $H + (n - 1)s_f$, where the prime stride s_f depends on frequency rather than on H : 127 for intraday/hourly, 27 for business-daily/daily, 11 for weekly, 5 for monthly, and 3 for quarterly data. The remaining earlier observations form the datastore.

Each task and range declares an alignment period, fixed and rolling strides, and a bounded retrieval lookback in the dataset configuration. Adaptation training and validation retain only rows with this complete retrieval history, so distances compare equal-width representations. Foundation forecasts remain separate: they use all history available at an origin, capped at the backbone limit (8192 for Chronos-2, 4096 for TS-ICL, and 2048 for Chronos-Bolt). TS-RAG retains its separate native length of 512 when used as an external control. Every official test window remains present.

For a query time t , let d_{max} be the fixed datastore endpoint and P the dataset period. The latest eligible neighbor date is

$$d_t = \min(t - H, d_{\text{max}}) - (\min(t - H, d_{\text{max}}) - t) \bmod P. \quad (1)$$

Earlier candidates walk backward from d_t by a datastore stride that is a multiple of P . This keeps the datastore fixed while aligning its phase to the query.

By default every eligible earlier datastore date is accessible. An optional global window cap is divided evenly across variates, rounded down, and retains the most recent dates. The fitting set has an analogous optional recent-window cap; global fitting and same-variate fitting are explicit alternatives. TS-RAG requires every variate to retain at least 11 datastore dates. Shared preparation records these counts; before TS-RAG representation extraction, its project-owned data adapter checks the minimum globally, including when a previously completed artifact is reused. Generic preparation remains available to the Ridge family, whose insufficient-history path is the vanilla fallback described below. An early training or validation origin that cannot supply the fixed history is excluded from fitting, but no official test origin is removed. Virtual $K = 0$ is always admissible and needs no fit. Each positive- K candidate must separately exceed the configured training and validation support thresholds; when no fitted candidate qualifies, TIME’s unchanged test windows use cached vanilla forecasts.

Let q_t be the finite fraction of a query history. The default gate requires $q_t \geq 0.8$ before retrieval. Datastore histories satisfy the same threshold, and their complete retrieved future must be finite. A query is RAG-eligible only when it has at least K_{max} valid chronological neighbors, where $K_{\text{max}} = 15$ for the default grid. Every candidate K uses this common query support and consumes the first K neighbors from the same ordered list; fewer than K_{max} valid neighbors makes the query ineligible for all candidates. After selection is frozen, test extraction retrieves only the selected K ; a test query needs those K neighbors rather than K_{max} .

2 Fixed ridge family

For each query, V is the vanilla foundation forecast and C is the forecast obtained when the K retrieved neighbor trajectories are supplied as covariates. For neighbor i , Y_i is its observed horizon and N_i is the foundation model’s vanilla forecast from that neighbor’s own history. The full design is

$$X = [V, C, Y_1, \dots, Y_K, N_1, \dots, N_K]. \quad (2)$$

The paired Y_i and N_i terms expose neighbor residual information. Let $s_{t,c}$ be the root mean squared seasonal difference over the complete history before query t for channel c . Only pairs whose two original dates are finite contribute, and their count is the denominator. For `full_ridge_shared`, one no-intercept ridge vector is shared across variates and forecast horizons and fitted by MSSE:

$$\hat{\beta}_\alpha = \arg \min_{\beta} \sum_{t,c,h} \left(\frac{(Y - V)_{t,c,h} - (X\beta)_{t,c,h}}{s_{t,c}} \right)^2 + \alpha \|\beta\|_2^2, \quad (3)$$

$$V' = V + X\hat{\beta}_\alpha. \quad (4)$$

The primary setting is $K = 10$, $\alpha = 10^{-2}$. Validation searches $K \in \{1, 5, 10, 15\}$ and $\alpha \in \{10^{-3}, 10^{-2}, 10^{-1}\}$ and compares every fitted method to virtual $K = 0$. The nested `cov_ridge_shared` design uses $[V, C]$; `y_ridge_shared` uses $[V, Y_1, \dots, Y_K]$; and `full_ridge_per_variate` retains the full design while fitting a separate coefficient vector from each variate’s training windows. All designs retain the outer vanilla term V in $V + X\beta$. Delta, convex, and native multivariate variants are outside the current protocol.

Validation first averages each candidate’s MSSE across variates at every date. A paired moving-date-block bootstrap estimates the standard error of each candidate’s loss difference from the observed best. The selected set is all candidates within one standard error; its deterministic preference is vanilla, then lower K , then larger α . The automatic block length is $\max(\text{round}(T^{1/3}), \lceil H/s_f \rceil)$ for T validation dates, capped below T ; 1000 replications are used by default.

Ridge fitting uses complete adaptation-training rows only; it never applies an element-wise NaN mean to the normal equations. During validation and test, an ineligible query receives the vanilla output for both C and V' . This gate uses only query and datastore information available at forecast time. Missing future labels affect metric support only and cannot select the prediction path.

3 Bayesian covariate baseline

For each candidate K , every eligible adaptation-training window is one paired trial. Let $z_i = 1$ when the per-window MSSE of C_i is lower than that of V_i , $z_i = 0$ when it is higher, and $z_i = 1/2$ for a tie. With a Beta(1, 1) prior, the posterior-mean probability is

$$p = \frac{1 + \sum_i z_i}{2 + n}. \quad (5)$$

Adaptation validation compares these frozen candidate mixtures by MSSE and selects their K , or the virtual vanilla candidate. The selected evidence is then frozen before test. The soft baseline is

$$B = (1 - p)V + pC. \quad (6)$$

An ineligible test window receives V for both the hard covariate and Bayesian branches. A task-level insufficient-training fallback fixes $p = 0$ and makes all family predictions equal to V .

A second Bayesian candidate calls Chronos-2 with every other variate supplied as a past-only covariate. The same paired training-win, Beta(1, 1) posterior, validation selection, and vanilla fallback logic mixes this forecast with V . No future covariate value is exposed.

4 Rolling per-variate horizon ridge

The independent `rolling_y_ridge_horizon` method has no validation selection. At every official test query it fits one coefficient vector per variate and horizon with design $[V, Y_1, \dots, Y_K]$. Defaults are $K = 15$ and $\alpha = 1$. Fitting

dates come only from the query variate, are aligned to the query’s dataset-period phase, retain at most the latest 100 dates, and require at least 64. Each fitting target must be fully observed before the test query.

Retrieval uses a causal cross-variate datastore at every fitting and query date. Its default 10,000-window cap is divided evenly across variates, and the smallest available balanced count fixes one common datastore size over all adapted dates. Incremental sufficient statistics remove expired fitting dates and add new ones as the test origin advances. A query with insufficient fitting dates or retrieval support returns V .

5 Evaluation

The official-test evaluations cover vanilla V , hard retrieval-context C , both Bayesian gates, the nested shared Ridge designs, shared and per-variate full Ridge, the validation-selected fixed family, rolling horizon Ridge, and the independently implemented TS-RAG control. Seasonal Naive supplies the task scaling baseline. Each method receives its own standard TIME artifact on identical official windows. The evaluator retains MSE, MAE, RMSE, MAPE, sMAPE, MASE, ND, and CRPS. Seasonal Naive remains the inherited external baseline used by broader TIME summaries. Missing dates are never removed or shifted: forecast errors skip non-finite target dates and MASE scales count only finite lagged pairs at their original timestamps. The comparison reports each metric mean together with its finite and total value counts; unequal finite coverage remains visible rather than preventing report construction. Conclusions require completed and inspected cluster artifacts.

Eligibility arrays, reason codes, effective training counts, and test RAG coverage, Ridge coefficients, Bayesian evidence, rolling fitting support, and cache identities are retained with the task artifacts. Vanilla forecasts and exact-context representations are shared across fixed and rolling methods only when prepared data, model, weights, source window, and context length agree; neighbor selections remain method-specific. The resulting adapted estimands are transparent gated hybrids: their selected method on eligible windows and vanilla otherwise. An insufficient-training task is the limiting case with zero eligible Adapttime windows and vanilla predictions for every fixed-family branch.